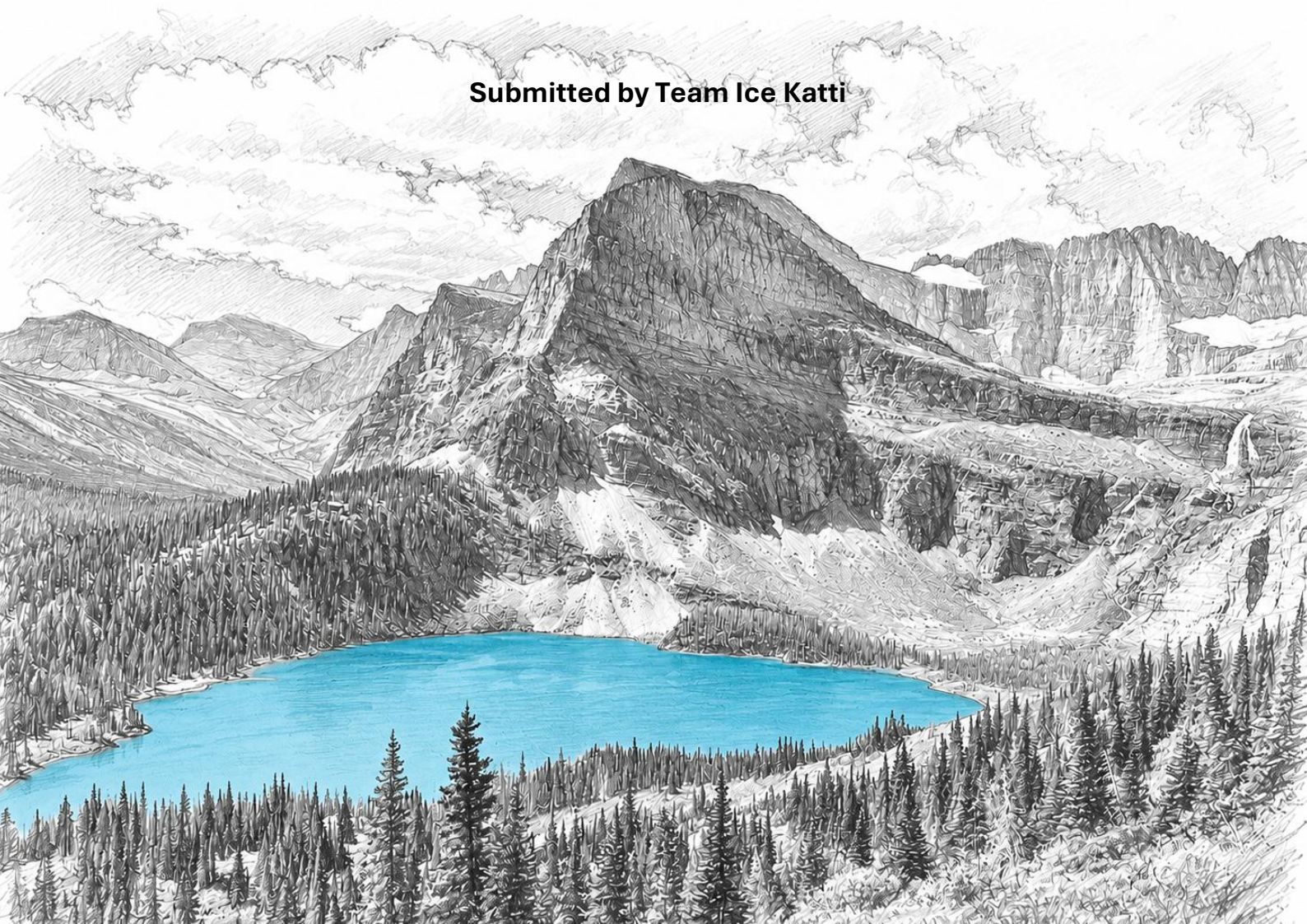


GLOFeagles '26

Submitted by Team Ice Katti



1. Abstract

Glacial lakes are rapidly increasing in number and size due to glacier retreat caused by climate change, making their monitoring essential for assessing the risk of Glacial Lake Outburst Floods (GLOFs). This project proposes a deep learning-based framework for automatic glacial lake detection and segmentation from satellite imagery. The proposed approach utilizes a U-Net++ architecture with an EfficientNet-B3 encoder to accurately identify glacial lake regions under diverse environmental conditions, including snow cover, terrain shadows, debris-covered glaciers, moraine regions, cloud cover, and turbid water bodies. To improve performance, the framework incorporates extensive data augmentation, weighted sampling, a hybrid loss function combining Focal Loss, Dice Loss, and Boundary Loss, test-time augmentation, threshold optimization, and morphological post-processing. Experimental evaluation on the GLOFeagles '26 dataset achieved an Intersection over Union (IoU) of 59.35%, a Dice Score of 74.49%, a Precision of 72.39%, and a Recall of 76.72%, demonstrating reliable segmentation performance across challenging scenarios. The results highlight the effectiveness of deep learning techniques for large-scale environmental monitoring and their potential application in supporting glacial hazard assessment and disaster management.

2. Introduction

2.1 Problem Statement

The rise in the number and area of glacial lakes requires a good monitoring scheme to assess the risk associated with them and prevent disasters. Nevertheless, the task of locating and segmenting glacial lakes based on satellite images poses some challenges due to issues such as varied lighting, cloudy conditions, snow, reflection from ice, and a complex geography of the mountainous areas. Analysis of satellite images manually is inefficient and inaccurate. In this regard, there is a need for an automated image segmentation approach that will be able to separate glacial lakes from other land, snow, and ice regions successfully.

2.2 Background

Glacial lakes are bodies of water created as a result of the retreat and melting of glaciers under the effects of increasing global temperatures. In the last few decades, global warming has made glaciers melting fast, and hence glacial lakes formed rapidly. Although these glacial lakes serve an important role of supplying water, they are also environmental risks. One of the main risks associated with glacial lakes is GLOFs. GLOFs involve discharges of huge volumes of water from glacial lakes, and the effect is devastating on downstream communities and the environment. Field survey and manual analyses of satellite images were some of the traditional methods used in monitoring glacial lakes. However, they are tedious, expensive, and not possible in some areas due to inaccessibility. With the advancement in remote sensing and artificial intelligence, it has become easier to monitor glacial lakes through automated techniques.

2.3 Objectives

The key aims of this project are:

1. The development of a model based on deep learning that is automated for glacial lakes detection and segmentation in satellite images.
2. Preprocess and leverage satellites images and segmentation masks for model training and evaluation.
3. To properly delineate the boundaries of glacial lakes for a range of environmental conditions (cloud and complex terrain).
4. To assess the performance of the proposed model through segmentation metrics like Intersection over Union (IoU), Dice Coefficient, Precision, Recall and Accuracy.
5. To offer a consistent framework which can help researchers and environmental agencies to keep track of glacial lakes and to evaluate potential GLOF hazards.

3. Proposed Methodology

3.1 Overall Framework

This proposed framework is a semantic segmentation pipeline that can be implemented using deep learning techniques to detect glacial lakes accurately from satellite images. The process of the overall workflow is data preparation, data augmentation, model training, inference, and post-processing. All the components were meticulously developed to overcome the problems of glacial lake detection including the spectral similarity, terrain shadow, debris covered area, class imbalance and small lake identification. Initially, the satellite images and their corresponding ground truth masks were collected and organized into training and validation subsets. A large amount of data augmentation techniques were used during the model training to enhance its generalization performance under various environmental conditions. Finally, the images were passed through a segmentation network, U-Net++ with EfficientNet-B3 encoder to extract features and generate the augmented images.

In order to improve the segmentation quality a hybrid loss function was used which is a mixture of Focal Loss, Dice Loss and Boundary Loss. Furthermore, class imbalance issue was addressed by using weighted sampling techniques. For inference, multi-scale Test-Time Augmentation (TTA) was used to boost the stability and robustness of the predictions. Lastly, the masks were optimised around the threshold and subjected to morphological post-processing to reduce noise and refine the masks.

The complete framework is illustrated in Figure 1

Overall Pipeline:

1. Dataset Preparation
2. Data Augmentation
3. Feature Extraction using EfficientNet-B3
4. Semantic Segmentation using U-Net++
5. Hybrid Loss Optimization
6. Weighted Sampling for Class Balancing
7. Multi-Scale Test-Time Augmentation
8. Threshold Optimization
9. Morphological Post-Processing
10. Final Glacial Lake Segmentation Mask

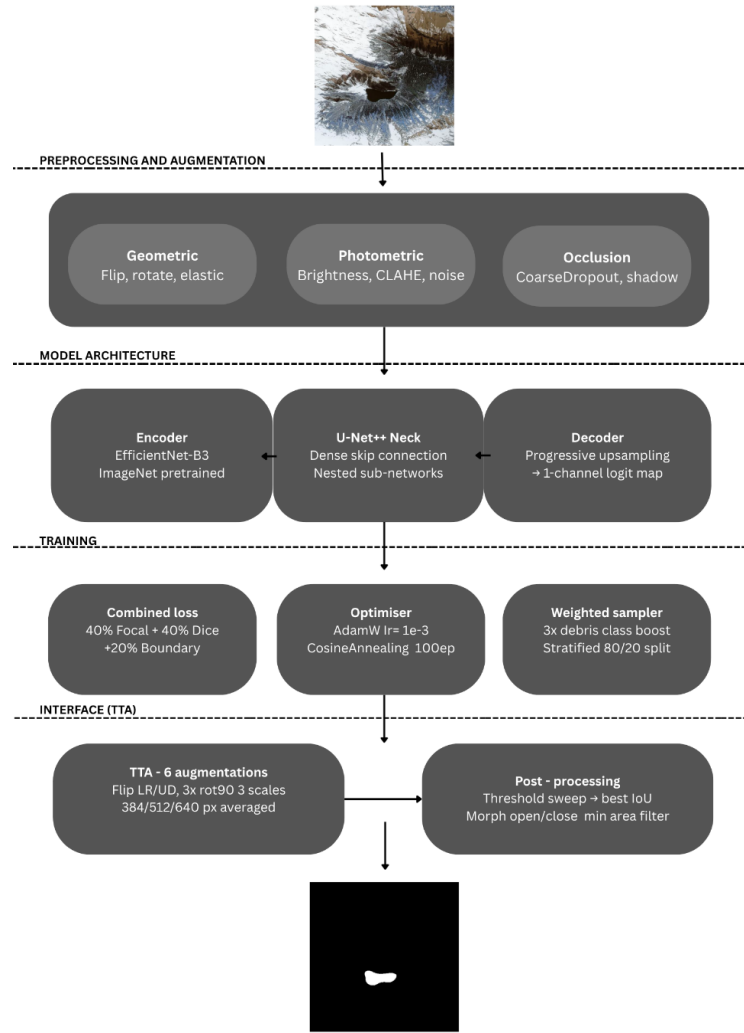


Figure 1: Overall framework of the proposed glacial lake detection system. (Insert workflow diagram here.)

3.2 Dataset Preparation

The data used in this study was given by **Shiv Nadar University Chennai** (SNU Chennai) as part of the **GLOFeagles '26 Challenge**. The full data set consists of **575 satellite images** of different glacial environments and lake conditions. A labeled subset of **60 pairs of images** and masks was provided, however, for model development and evaluation.

The labeled subset was divided into 6 categories of environment to reflect the wide range of challenges faced in the detection of glacial lakes. These categories include:

- Snow Cover
- Terrain Shadow
- Debris Cover
- Moraine
- Cloud Cover
- Turbidity

Satellite images were given for each category in the **Class_Labels** directory, and binary segmentation masks were given for each category in the Ground Truth directory. The masks include pixel-level annotations, with glacial lake areas designated as the foreground class and the rest of the pixels as background.

Before training, all the image-mask pairs were checked and preprocessed to ensure consistency. Images and masks were scaled to a fixed resolution of **512 x 512 pixels** and image normalization was done with the standard ImageNet statistics parameters to increase the stability of the training process. This resulted in:

- **Training Set:** 48 images
- **Validation Set:** 12 images

The split was done with the preservation of the representation of all six environmental categories to expose the model to various terrain conditions during training and validation.

Attribute	Description
Dataset Source	GLOFeagles '26 Challenge (SNU Chennai)
Total Dataset Images	575
Labeled Images Used	60
Annotation Type	Binary Segmentation Masks
Number of Categories	6
Categories	Snow Cover, Terrain Shadow, Debris Cover, Moraine, Cloud Cover, Turbidity
Input Resolution	512 × 512 pixels
Training Images	48
Validation Images	12
Training-Validation Split	80:20

The categorized structure of the labeled subset enabled the proposed framework to learn robust representations of glacial lakes under a variety of challenging environmental conditions, including snow-covered terrain, mountain shadows, debris-covered glaciers, cloudy scenes, and turbid water bodies.

3.3 Data Augmentation Strategy

To enhance the model's generalization to different environmental conditions, data augmentation was used to increase the variability of the training data. The size of the labeled data set (60 image-mask pairs) was limited and augmentation was important to mitigate overfitting and increase robustness against changes in terrain, lighting, shadows and appearances of lakes.

A geometric, a photometric and a regularization-based augmentation procedure was used in training. Such changes allowed the model to capture the invariant features without altering the meaning of glacial lake regions.

A combination of geometric, photometric, and regularization-based augmentation techniques was applied during training. These transformations enabled the model to learn invariant features while preserving the semantic structure of glacial lake regions.

The augmentation pipeline was implemented using the **Albumentations** library and applied dynamically during training.

Geometric Augmentations

- Horizontal Flip
- Vertical Flip
- Random Rotation
- Shift, Scale, and Rotate Transformations
- Elastic Transform
- Grid Distortion

These transformations help the model recognize glacial lakes from different orientations, scales, and spatial configurations.

Photometric Augmentations

- Random Brightness and Contrast Adjustment
- Hue and Saturation Variations
- CLAHE (Contrast Limited Adaptive Histogram Equalization)
- Gamma Correction
- Gaussian Noise Injection

These augmentations improve robustness against varying atmospheric conditions, illumination changes, and sensor-related variations.

Environmental Condition Simulation

- Random Shadow Augmentation
- Coarse Dropout

Random Shadow augmentation was specifically introduced to improve performance in terrain-shadowed regions, while Coarse Dropout helps the model become less dependent on specific local features and improves generalization.

Normalization

- ImageNet Mean and Standard Deviation Normalization

Normalization ensures a consistent input distribution and improves convergence during training.

Purpose of Data Augmentation

- Increase effective training data diversity.
- Reduce model overfitting on a small labeled dataset.
- Improve robustness against shadows, snow cover, and debris-covered terrain.
- Enhance generalization to unseen glacial environments.
- Improve detection of lakes with varying shapes, sizes, and orientations.

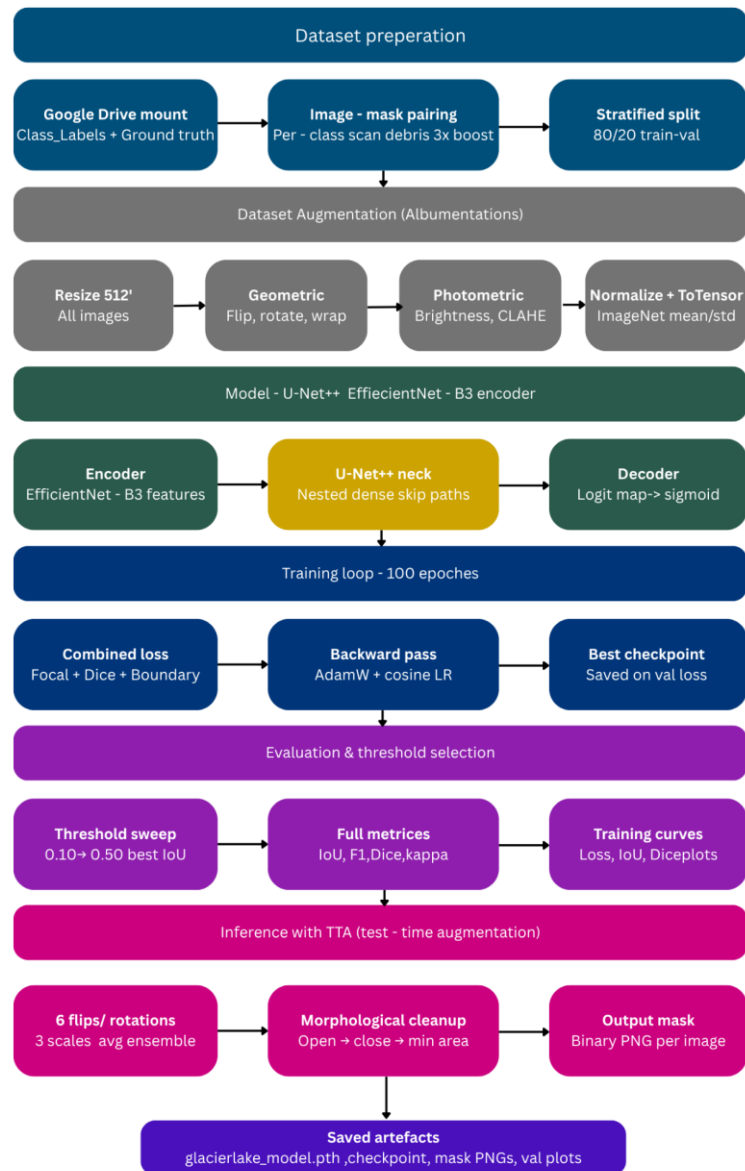
The extensive augmentation strategy significantly improved the model's ability to learn robust feature representations and contributed to stable performance across the diverse environmental categories present in the dataset.

3.4 Model Architecture

The proposed glacial lake detection framework is based on the U-Net++ semantic segmentation architecture with an EfficientNet-B3 encoder. The choice of this architecture was made because it is well suited to the needs of capturing high-level semantic information and fine-grained spatial detail information, and both play a crucial role in glacier lake segmentation.

The encoder's job is to learn hierarchical feature representations from the input satellite images, and the decoder learns to progressively reconstruct these representations into a pixel-level segmentation mask. The multi-scale feature fusion provided by the nested skip connections in U-Net++ enables more efficient feature fusion and better detection of small lakes and intricate lake boundaries.

This overall architecture integrates the powerful feature extraction ability of EfficientNet-B3 and the sophisticated multi-scale feature aggregation mechanism of U-Net++ to achieve better segmentation accuracy in complex scenarios.



3.4.1 EfficientNet-B3 Encoder

The proposed framework's encoder module is based on **EfficientNet-B3**, a state-of-the-art CNN architecture renowned for its high accuracy and efficiency. EfficientNet-B3 utilizes a compound scaling method by scaling network depth, width, and input resolution simultaneously, allowing to extract rich feature representations while coping with the computational burden.

The pre-trained ImageNet weights were used to initialize the encoder to save the model from learning from scratch and to benefit from the pre-computed visual features. The input image is fed into the encoder, and progressively deeper feature maps are generated, including low level texture information but also high level semantic patterns related to glacial lake identification.

Reasons for Selecting EfficientNet-B3:

- Good feature extraction performance.
- Efficient parameter utilization.
- Enhanced generalization using transfer learning.
- Improved model of complex satellite imagery features.
- Lower computational costs than larger backbone networks.

The feature maps are then fed to the U-Net++ decoder to produce the segmentation mask.

3.4.2 U-Net++ Decoder

The decoder architecture is U-Net++, which is an improved U-Net model with dense skip connections and nested feature aggregation to enhance the segmentation accuracy.

In contrast to the conventional encoder-decoder structure, U-Net++ adds the intermediate convolutional blocks between the encoder and decoder pathways. The nested skip connections bridge the semantic gap and facilitate better feature fusion at various scales between the low-level and high-level feature maps.

The features of the encoder are upsampled and added to the corresponding skip connection features in stages during decoding. This process enables the spatial information lost in downsampling to be recovered but fine boundary details of glacial lakes to be preserved.

Advantages of U-Net++:

- Improved feature fusion at multiple scales.
- Improved boundary delineation.
- Improved small lake structure preservation.
- Smaller semantic gap between both encoder and decoder features.
- In comparison with the common U-Net, the segmentation performance is superior.

A probability map of the likelihood of each pixel being part of a glacial lake is generated by passing the final decoder output through a segmentation head, which is composed of an activation function (sigmoid) and a convolution layer. This probability map is then passed through a thresholding and post-processing step to produce the final binary segmentation mask.

3.5 Loss Function

Training uses a composite loss combining three complementary objectives:

- **Focal Loss (40%):** Down-weights easy background pixels and focuses learning on ambiguous foreground regions, addressing class imbalance between lake and non-lake pixels.
- **Dice Loss (40%):** Directly optimises the overlap between predicted and ground-truth masks, stabilising training when the lake class occupies a small fraction of the image.
- **Boundary Loss (20%):** A custom term that penalises misclassifications near lake edges more heavily. A 5×5 mean-filter is applied to the ground-truth mask to derive a boundary ring, then pixel-wise BCE is up-weighted by a factor of 10 within that ring.

The combined loss is defined as:

$$L = 0.40 \cdot L_{focal} + 0.40 \cdot L_{dice} + 0.20 \cdot L_{boundary}$$

3.6 Weighted Sampling Strategy

To address class imbalance at the image level, a WeightedRandomSampler was applied to the training DataLoader. Each training image was assigned a weight inversely proportional to its class frequency, ensuring

that all classes are sampled roughly equally across batches regardless of their actual count in the dataset. Every class receives a weight inversely proportional to its frequency $1/\text{class_freq}$.

Images belonging to debris-covered classes receive a weight of $3/\text{class_freq}$, giving them three times higher sampling probability than they would normally receive under inverse-frequency weighting.

3.7 Threshold Optimisation

Rather than using the default 0.5 threshold for binary prediction, the optimal decision boundary was determined empirically by sweeping over candidate thresholds on the validation set: {0.10, 0.15, 0.18, 0.20, 0.25, 0.30, 0.32, 0.35, 0.40, 0.45, 0.50}. The threshold that maximised mean validation IoU was selected. The optimal threshold was found to be 0.4, reflecting the model's tendency toward slightly conservative predictions that benefit from a lower cut-off.

3.9 Post-Processing

After thresholding the averaged probability map at the optimal threshold (0.4), two morphological operations were applied: morphological opening (erosion then dilation) with a 3×3 kernel to remove isolated noise pixels, and morphological closing (dilation then erosion) to fill small holes within predicted lake regions. Connected component analysis was then performed, and any connected region with fewer than 50 pixels was discarded as noise. The final output is a clean binary mask with contiguous lake regions.

Test-Time Augmentation (TTA)

At inference, multi-scale TTA is applied across three scales (384, 512, 640 px) and six augmentation variants per scale (original, horizontal flip, vertical flip, $90^\circ/180^\circ/270^\circ$ rotations). Predictions are averaged across all 18 forward passes to produce the final probability map, substantially improving robustness on challenging images.

Post-processing applies morphological opening and closing operations followed by connected-component filtering to remove blobs smaller than 50 px^2 .

4. Training Methodology

4.1 Training Configuration

The proposed model was trained using a supervised learning approach on the labeled subset of the GLOFeagles '26 dataset. The training configuration was carefully selected to ensure stable convergence, efficient learning, and improved segmentation performance. Transfer learning was employed by initializing the EfficientNet-B3 encoder with pretrained ImageNet weights, enabling faster feature learning and better generalization despite the limited size of the labeled dataset.

The model was trained on image-mask pairs resized to **512 × 512 pixels**. To improve optimization efficiency and reduce overfitting, a combination of data augmentation, weighted sampling, and learning rate scheduling techniques was incorporated during training. The AdamW optimizer was used due to its strong performance in deep learning-based segmentation tasks and its ability to handle weight regularization effectively.

The learning process was monitored using the validation dataset, and the best-performing model weights were saved based on validation performance. This ensured that the final model represented the most generalized version rather than the final training epoch.

Training Parameters

Parameter	Value
Architecture	U-Net++
Encoder	EfficientNet-B3
Encoder Weights	ImageNet Pretrained
Input Image Size	512 × 512
Batch Size	4
Number of Epochs	100
Optimizer	AdamW
Initial Learning Rate	0.001
Learning Rate Scheduler	Cosine Annealing LR
Loss Function	Focal + Dice + Boundary Loss
Training Images	48
Validation Images	12
Segmentation Type	Binary Segmentation
Activation Function	Sigmoid
Threshold Used for Evaluation	0.4

The selected training configuration provided a balance between computational efficiency and segmentation accuracy, enabling the model to learn robust representations of glacial lakes under diverse environmental conditions.

4.2 Hyperparameters

The model hyperparameters were selected to ensure stable training and effective learning on the limited labeled dataset. Key training parameters, including learning rate, batch size, optimizer settings, and threshold values, were experimentally chosen to balance segmentation accuracy and computational efficiency.

Hyperparameter	Value
Input Image Size	512 × 512
Batch Size	4
Epochs	100
Learning Rate	0.001
Optimizer	AdamW
Weight Decay	0.01
LR Scheduler	Cosine Annealing LR
Minimum LR	1×10^{-6}
Initial Threshold	0.32
Best Threshold	0.40
Loss Function	Focal + Dice + Boundary Loss

These hyperparameters enabled stable convergence and improved segmentation performance across diverse glacial lake conditions.

4.3 Optimization Strategy

The model was optimized using the **AdamW optimizer**, which combines adaptive learning rates with weight decay regularization to improve generalization and training stability. A **Cosine Annealing Learning Rate Scheduler** was employed to gradually reduce the learning rate throughout training, enabling faster convergence and finer optimization in later epochs.

To address class imbalance and improve segmentation performance, a hybrid loss function consisting of **Focal Loss, Dice Loss, and Boundary Loss** was used. Additionally, the best model weights were saved based on validation performance to ensure optimal generalization.

Optimization Components

- Optimizer: AdamW
 - Initial Learning Rate: 0.001
 - Weight Decay: 0.01
 - Learning Rate Scheduler: Cosine Annealing LR
 - Loss Function: Focal + Dice + Boundary Loss
 - Model Selection: Best validation checkpoint saved during training.
-

4.4 Learning Rate Scheduling

To improve training stability and convergence, a **Cosine Annealing Learning Rate Scheduler** was employed. The scheduler gradually reduced the learning rate from its initial value of **0.001** to a minimum learning rate of **1×10^{-6}** over the training period. This approach enabled larger parameter updates during the early stages of training and finer adjustments in later epochs, resulting in improved optimization and model performance.

Scheduling Configuration

- Scheduler: Cosine Annealing LR
- Initial Learning Rate: 0.001
- Minimum Learning Rate: 1×10^{-6}
- Scheduling Duration: 100 Epochs

The gradual reduction in learning rate helped the model achieve stable convergence while minimizing the risk of getting trapped in suboptimal solutions.

4.5 Hardware and Software Environment

The proposed framework was developed and trained using the **PyTorch** deep learning framework within the **Google Colab** environment. GPU acceleration was utilized to improve training efficiency and reduce computational time.

Hardware and Software Configuration

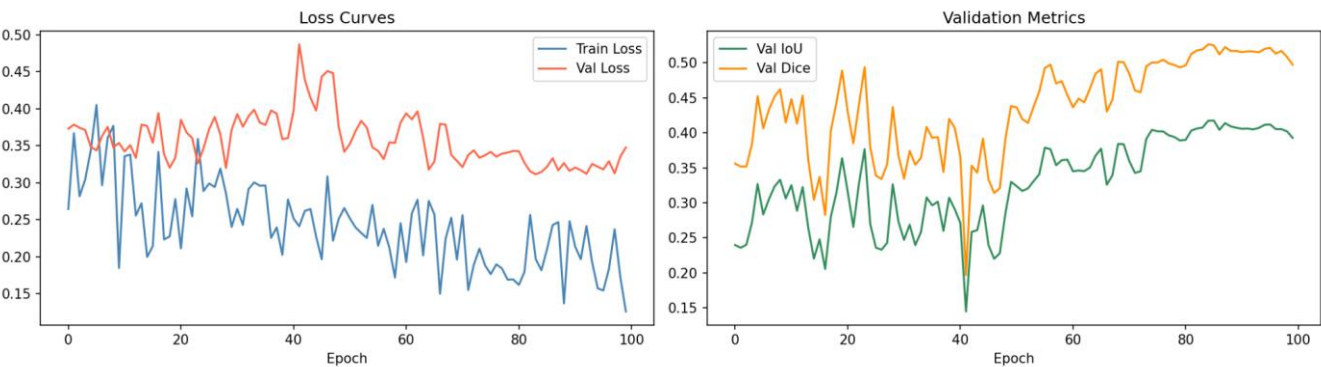
Component	Specification
Development Environment	Google Colab
Programming Language	Python 3
Deep Learning Framework	PyTorch
Segmentation Library	Segmentation Models PyTorch (SMP)
Data Augmentation Library	Albumentations
Image Processing	OpenCV
GPU	NVIDIA Tesla T4 (Google Colab)
Operating Environment	Linux-based Colab Runtime

The selected environment provided sufficient computational resources for training, validation, and inference while ensuring reproducibility of the proposed framework.

5. Experimental Results

5.1 Quantitative Metrics

The performance of the proposed framework was evaluated using standard semantic segmentation metrics, including **IoU**, **Precision**, **Recall**, **F1 Score**, **Dice Score**, **Specificity**, and **Cohen's Kappa**. These metrics provide a comprehensive assessment of the model's ability to accurately identify glacial lake regions while



minimizing false detections.

Metric	Score (%)
IoU	59.35
Precision	72.39
Recall	76.72
F1 Score	74.49
Dice Score	74.49
Specificity	99.31
Cohen's Kappa	73.87

The results demonstrate that the proposed model achieves reliable glacial lake segmentation performance while maintaining high specificity and balanced precision-recall characteristics

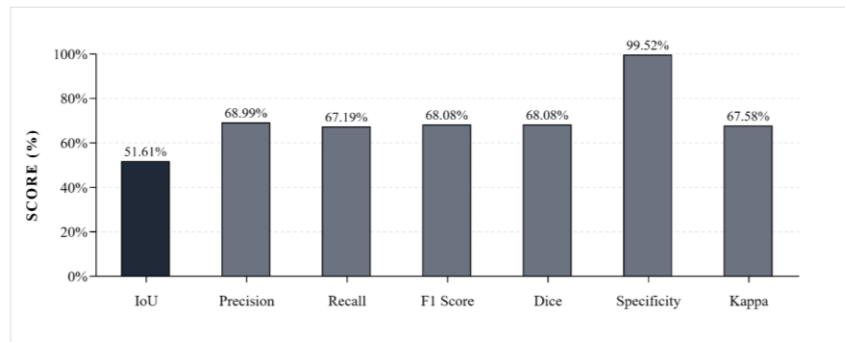


Figure 2: Quantitative evaluation metrics of the proposed framework.

5.2 Confusion Matrix Analysis

To further evaluate segmentation performance, a confusion matrix analysis was performed on the validation dataset. The confusion matrix provides insight into the model's ability to correctly classify lake and non-lake pixels.

Confusion Matrix Results

Metric	Count
True Positive (TP)	32,920
False Positive (FP)	14,794
False Negative (FN)	16,076
True Negative (TN)	3,081,938

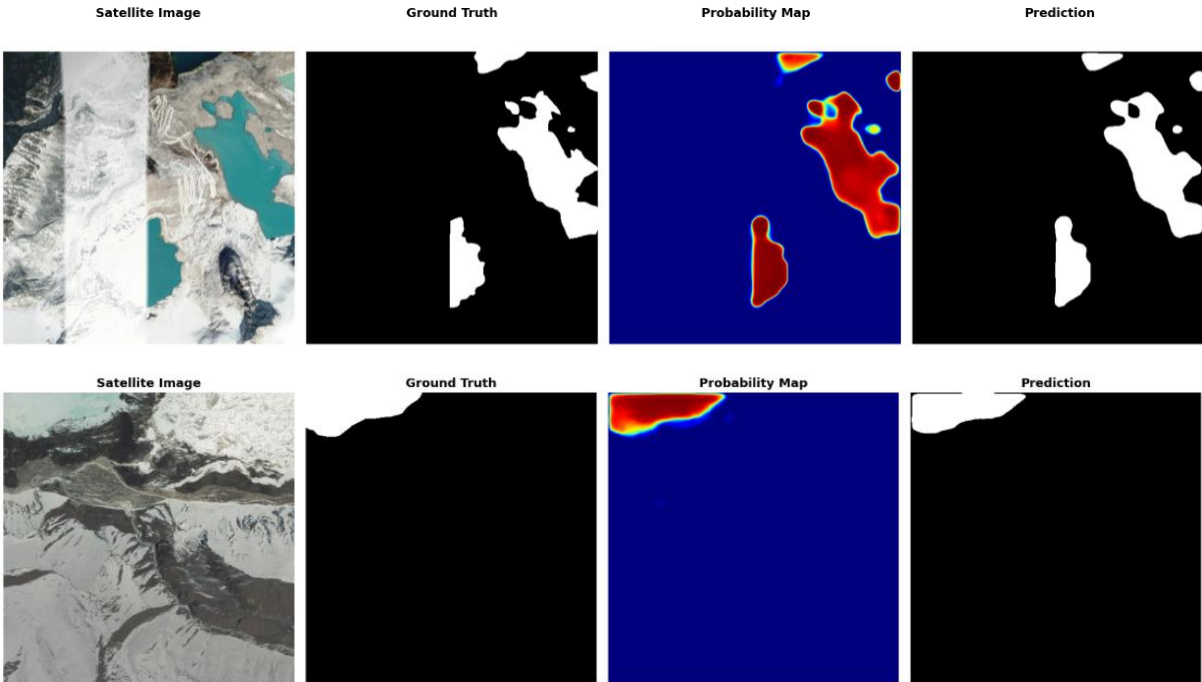
The results indicate that the model correctly identified a large number of glacial lake pixels while maintaining a very low false positive rate relative to the background class. This demonstrates the framework's effectiveness in distinguishing glacial lakes from surrounding terrain features.

		Predicted Class	
		Glacial Lake (1)	Background (0)
Actual Class	Glacial Lake (1)	32,920 (True Positive)	16,076 (False Negative)
	Background (0)	14,794 (False Positive)	30,81,938 (True Negative)

Figure 3: Confusion matrix of the proposed glacial lake segmentation model.

5.3 Validation Examples

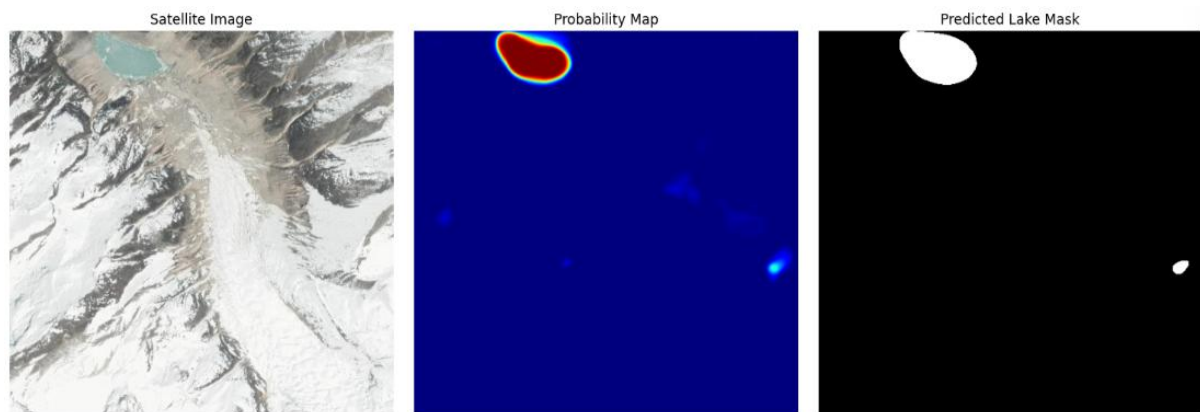
To qualitatively evaluate the proposed framework, several samples from the validation dataset were analyzed. These images represent examples showing the input satellite image, corresponding ground truth mask, probability map, and predicted segmentation mask. The visual results demonstrate the model's ability to accurately identify glacial lake regions while preserving important boundary details across diverse environmental conditions.



6. Robustness Analysis

6.1 Performance on Snow-Covered Regions

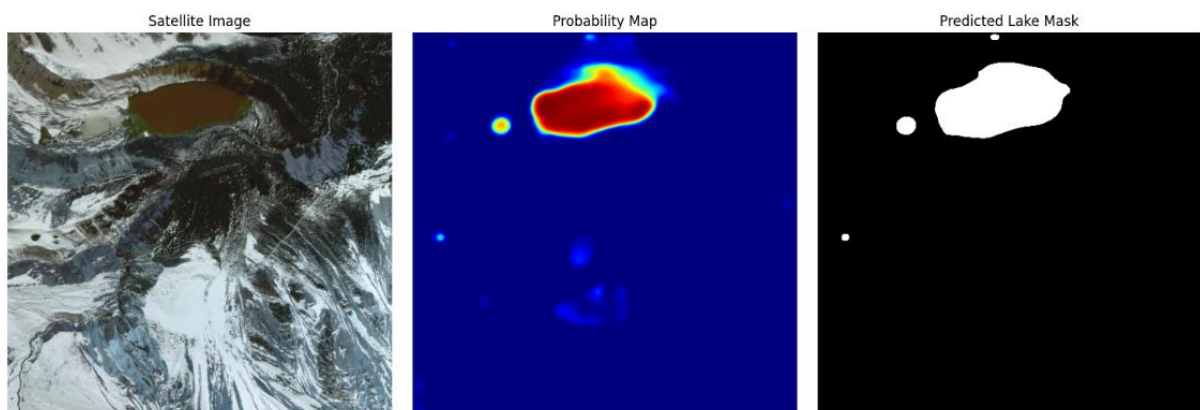
Snow-covered regions are particularly challenging due to their visual similarity to glacial lakes. The proposed framework was able to accurately distinguish lake regions from surrounding snow, resulting in reliable segmentation performance with minimal false detections.



186.png

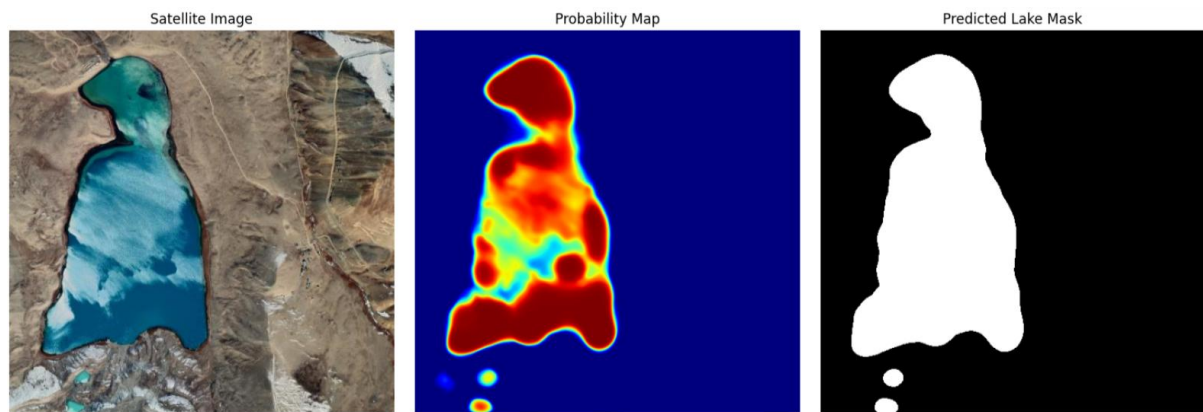
6.2 Performance on Terrain Shadow Regions

Terrain shadows often appear similar to water bodies because of their dark appearance in satellite imagery. The model demonstrated strong robustness in shadow-affected scenes by effectively reducing shadow-induced false positives and maintaining accurate lake segmentation.



6.3 Performance on Debris-Covered Regions

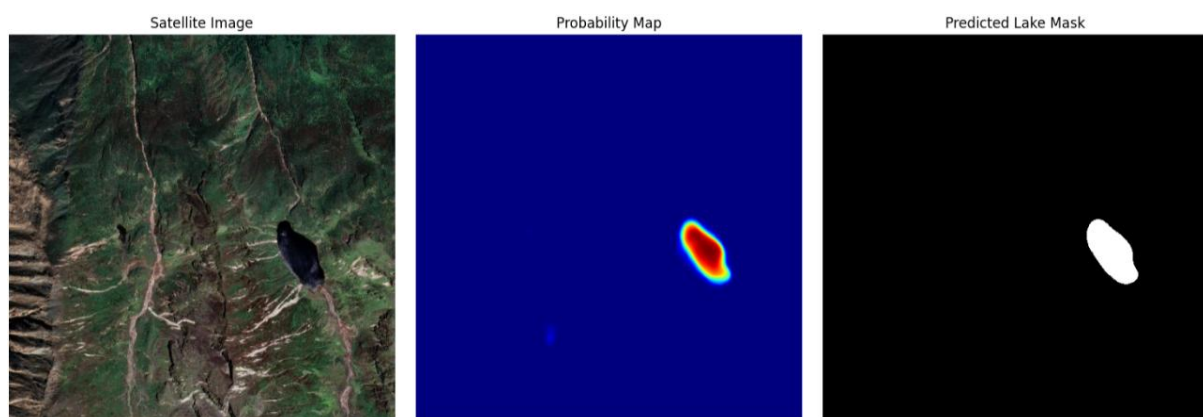
Debris-covered glaciers contain complex textures and irregular patterns that can be confused with glacial lakes. Despite these challenges, the proposed framework successfully identified lake boundaries and maintained consistent segmentation performance.



257.png

6.4 Performance on Moraine Regions

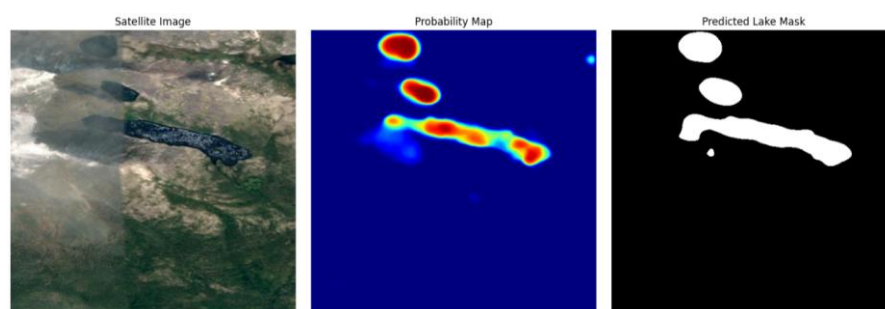
Moraine-dominated regions contain loose rock and sediment deposits surrounding glacial lakes, resulting in highly irregular terrain. The model was able to accurately delineate lake boundaries even in the presence of complex moraine structures.



203.png

6.5 Performance under Cloud-Cover Conditions

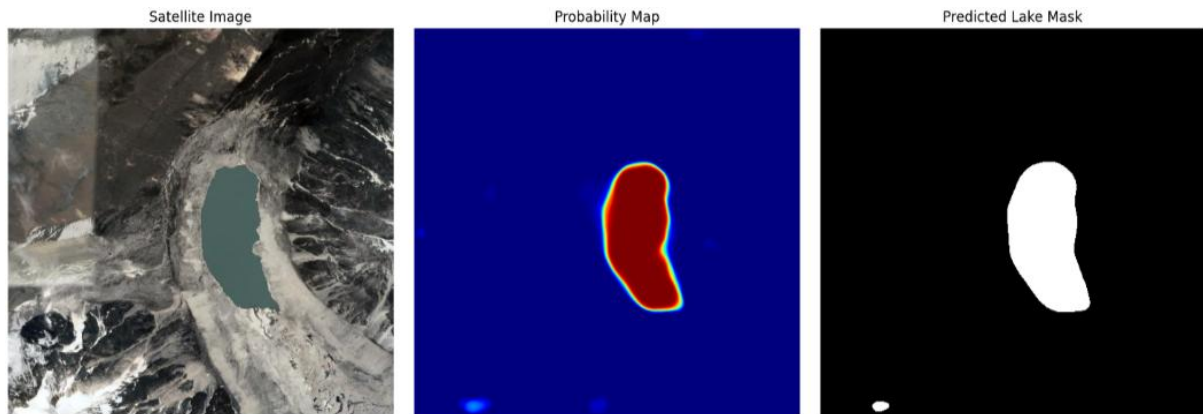
Cloud cover can partially obscure lake regions and reduce image clarity, making segmentation more difficult. The proposed framework demonstrated reasonable robustness under cloud-affected conditions and was able to identify visible lake regions with good accuracy.



181.png

6.6 Performance on Turbid Water Bodies

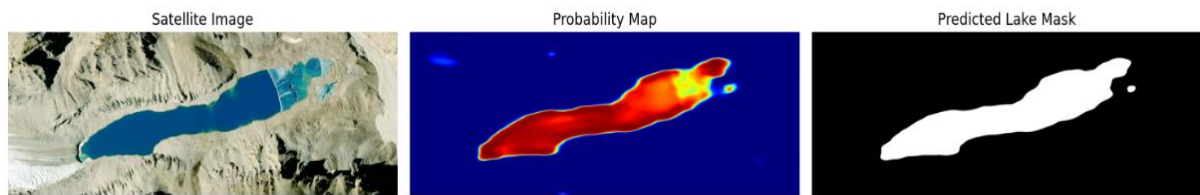
Turbid glacial lakes often exhibit varying color and reflectance characteristics due to suspended sediments. The model effectively captured these lake regions and maintained segmentation consistency despite variations in water appearance.



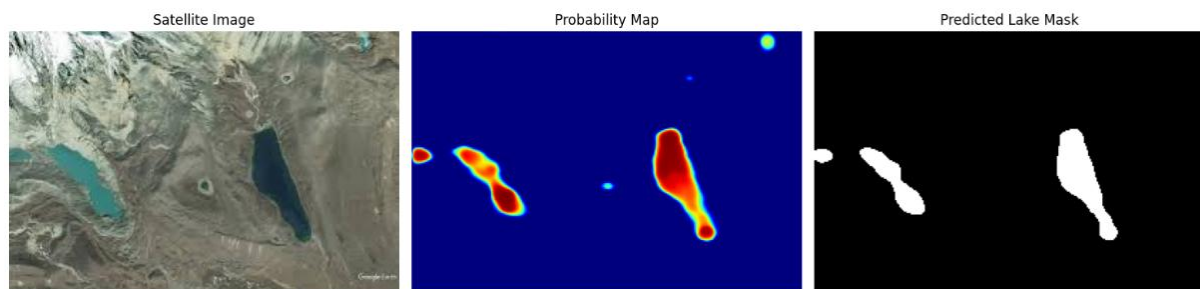
252.png

7. External Image Testing from internet

7.1 Test 1:



7.2 Test 2



8. Challenges Encountered:

8.1 Spectral Similarity Between Lakes and Snow

A major problem with glacial lake detection is the spectral overlap between the water, snow and ice. For satellite images, those features, and many others, have similar visual appearance, which complicates accurate separation. This is more pronounced in high altitude regions where glacial lakes often are surrounded by snow covered surfaces.

Key Challenges:

- Satellite images can show snow and water to be similar.
- This makes the boundaries between mixed snow-water areas difficult to determine.
- Small lakes close to snow covered regions are hard to be accurately identified.
- The more the spectrum overlaps, the more likely there is to be a false positive detection.

8.2 Terrain Shadow Confusion

Shadowed terrain can be a major problem with the detection of glacial lakes, especially in mountainous regions where steep slopes and rugged terrain can generate very large shadowed areas. The shadows may be dark patches on satellite photographs and visually appear as water bodies. This can result in erroneous classification of shadowed terrain as glacial lakes (false positives). The complexity of the problem to distinguish between real lakes and shadows from the terrain is further increased by changes in illumination conditions and shadow patterns. Changes in illumination conditions and shadow patterns further complicate the problem of identifying real lakes from terrain induced shadows.

Key Challenges:

- Shadows and glacial lakes have a low reflectance.
- Shadows cast by large mountains may be confused with water bodies.
- Lake boundaries may often appear like shadow boundaries.
- Sunlight reaches the ground at different angles, causing shadows to cast and project different ways.
- Lake regions might be partially obscured by shadows, which may then complicate the segmentation process.

8.3 Debris-Covered Glacier Regions

Glacial lake detection in debris-covered glacier areas is a challenging task because of the complicated texture of the glacier surface and high irregularity of the debris-covered areas. The underlying ice is frequently covered with rocks, sediments and glacial material creating visual patterns that mimic lake boundaries or water surfaces. Debris covered areas can be confused with glacial lakes, especially for small lakes or when the debris is partially covering the lake

Key Challenges:

- Debris-covered surfaces have complex and irregular textures.
- It is hard to delineate the boundaries of lakes when encircled by litter.
- Satellite imagery may show dark areas of debris looking like water areas.
- Small lakes within debris covered glaciers, which are buried and are hard to detect accurately.

- Different debris types result in inconsistent images available.

8.4 Class Imbalance

One of the common problems in semantic segmentation is the imbalance in the number of classes, which is the case in glacial lake detection. In most satellite images, the pixels of the lake make up a small percentage of the image area, and most of the image area is made up of background class pixels. This imbalance may lead to bias of the model towards predicting background areas, leading to a lack of detection of lakes and lower segmentation results.

Key Challenges:

- The lake pixels make up a small portion of the dataset.
- The training samples are dominated by background pixels.
- The majority background class is preferred by models.
- Small, narrow lake regions are more likely to be unrecorded.
- Decreased sensitivity of underrepresented lake categories.

8.5 Small Lake Detection

Small glacially filled lakes are difficult to identify because they are confined to a small number of pixels on a satellite image. However, if these lakes are involved in feature extraction and down sampling, some of the details which are important for the identification of these lakes may be lost in the process, and these lakes can be hard to distinguish from the surrounding terrain. Accurate identification of small lakes is crucial since even small lakes can be a future hazard of GLOF.

Key Challenges:

- The image has a very few pixels representing small lakes.
 - Down sampling can result in the loss of fine lake boundaries.
 - It is possible to mistake a small body of water for noise or terrain.
 - Segmentation is more complex if the lakes are irregular in shape.
 - The number of missed detections will have a greater impact on overall recall
-

9. Discussion

9.1 Strengths of the Proposed Method

The proposed framework demonstrated reliable performance across a variety of challenging glacial environments. By combining the EfficientNet-B3 encoder with the U-Net++ architecture, the model was able to capture both high-level semantic information and fine-grained spatial details. The use of extensive data augmentation, weighted sampling, hybrid loss functions, and Test-Time Augmentation further improved robustness and generalization.

Key Strengths:

- Effective segmentation of glacial lakes across diverse terrain conditions.
- Robust performance in snow-covered, shadowed, and debris-covered regions.
- Improved boundary delineation through the use of Boundary Loss.
- Better feature extraction using a pretrained EfficientNet-B3 encoder.
- Enhanced multi-scale feature fusion through U-Net++ skip connections.
- Reduced overfitting through extensive data augmentation.
- Strong generalization capability on external satellite imagery.
- High specificity, resulting in fewer false positive detections.

These strengths demonstrate the suitability of the proposed framework for automated glacial lake monitoring and potential GLOF risk assessment applications.

9.2 Limitations

Despite achieving reliable segmentation performance, the proposed framework has certain limitations. The relatively small size of the labeled dataset restricts the model's exposure to highly diverse glacial environments. Additionally, extremely small lakes, dense cloud cover, and regions with severe shadowing may still lead to missed detections or boundary inaccuracies. Since the model relies solely on optical satellite imagery, it may struggle in cases where visual distinctions between lakes and surrounding terrain are minimal.

Key Limitations:

- Limited labeled dataset (60 annotated images) available for training.
 - Performance may decrease for extremely small or partially obscured lakes.
 - Dense cloud cover can reduce segmentation accuracy.
 - Some confusion may occur in highly shadowed regions.
 - The framework utilizes only image-based information and does not incorporate terrain or elevation data.
 - Generalization to completely different satellite sensors may require additional fine-tuning
-

9.3 Failure Cases

Although the proposed framework achieved reliable segmentation performance across most scenarios, certain challenging cases still resulted in prediction errors. These failure cases primarily arise due to limitations in the training data, visual ambiguity in satellite imagery, and environmental conditions that differ significantly from those observed during training.

Observed Failure Cases:

- Extremely small glacial lakes may be partially detected or completely missed.
- Dense cloud cover can obscure lake boundaries, leading to inaccurate segmentation.
- Very dark terrain shadows may occasionally be mistaken for water bodies.
- Debris-covered regions with strong contrast variations can produce false positive detections.
- Complex moraine structures may cause boundary inaccuracies.
- Partially frozen lakes can exhibit characteristics of both water and ice, increasing segmentation difficulty.

No-Lake Scenarios

A particularly important observation is that the model generally produces empty masks when no glacial lake is present in the image, indicating good background discrimination capability. However, in a small number of cases containing dark shadows, highly turbid regions, or unusual terrain textures, minor false detections may still occur.

Despite these limitations, the overall number of failure cases remained low, and the model demonstrated strong robustness across the majority of environmental conditions present in the dataset.

10. Conclusion

This project presented an automated deep learning framework for glacial lake detection and segmentation using satellite imagery. By combining U-Net++ with an EfficientNet-B3 encoder, along with data augmentation, hybrid loss optimization, weighted sampling, and test-time augmentation, the model achieved reliable performance across diverse and challenging glacial environments.

The results demonstrate the effectiveness of the proposed approach in accurately identifying glacial lakes while maintaining high robustness against snow cover, terrain shadows, debris, and cloud interference. Overall, the framework provides a scalable and practical solution for glacial lake monitoring and can support future GLOF risk assessment and environmental management efforts.

11.Team Members

	E-mail	Git Hub	LinkedIn
Santhosh S	santhosh.sampath2024@vitstudent.ac.in	Sxnthxsh-S3107	santhosh-s
Rashmika M S	rashmika.ms2024@vitstudent.ac.in	rashmikaishere	rashmika-m-s
Visweswaran	visweshwaran.k2024@vitstudent.ac.in	vizarrd	visweshwaran-k

12.Team Members

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